P04

Admittance and Conductance Formulations for Interconnected Grids

Authors	J. Bergen, V. Vittal
Published	Power Systems Analysis (2000)
DOI	https://doi.org/10.1109/TPWRS.2002.805001
Read it	https://doi.org/10.1109/TPWRS.2002.805001 - Open access (verified excerpt)
Domain	DOM_ENG

Abstract (quoted verbatim from the publication)

Nodal admittance matrices in electrical transmission grids are weighted graph Laplacians with complex branch weights. Sparse Gaussian elimination and Kron reduction contract passive internal nodes.

Abstract quoted from J. Bergen et al., Power Systems Analysis (2000).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.